Gaurav Kumar Thakur

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Education

Maharishi Markandeshwar (Deemed to be) University — **Aug 2024 – Expected May 2028**
B.Tech, Computer Science and Engineering — CGPA: 7.53/10 — *Ambala, Haryana*

Experience

Software Engineer Intern · *FamilyShell LLP* — **May 2026 – Aug 2026**

- Redesigned and built the Family Tree module end to end (Flutter app, web UI, **FastAPI**, MongoDB), replacing one global shared graph with isolated per-tree workspaces and embedded adjacency lists; cut tree-load latency from **4.7 s to ~1 s (~79%)**.
- Ensured data integrity with **MongoDB transactions** for bilateral relationship sync, **optimistic concurrency control** (versioned documents), and a **validation engine** (cycle detection, age-gap and marriage rules), plus a soft-delete lifecycle.
- Built three **Redis cache layers**, owner/viewer **RBAC**, invitations, real-time sockets, and Flutter views (**BLoC**, genealogy_chart).

Open Source Contributions

Kubernetes – client-go (connrotation) · *Go, client-go* — PR #143105 · merged

- Fixed a **nil-pointer panic** where NewDialer and NewDialerWithTracker stored a nil dial function despite the documented default; now falls back to **net.Dialer.DialContext**, with a regression test for both constructors (also run under -race).

Kubernetes SIGs – DRA Example Driver · *Go, CDI* — PR #259 · merged

- Fixed DRA recovery after node or runtime restarts by recreating missing claim-specific **CDI spec** files when restoring prepared claims from checkpoint; added tests for missing specs, idempotent restore, and recreation failure.

Apache Beam – Dask Runner · *Python, Dask* — PR #40496 · merged

- Added an opt-in **--dask_lazy_side_inputs** mode that streams side-input bags one partition at a time (memory bounded by the largest partition) and uses **worker_client()** to avoid worker deadlocks; wrote 9 unit and distributed LocalCluster tests.

Apache Beam – Python Typehints · *Python, Type Hints* — PR #40625 · merged

- Fixed string annotations and forward references degrading to Any by resolving via **typing.get_type_hints**; added tests.

Apache Beam – Retry Utilities · *Python, BigQuery I/O* — PR #40631 · merged

- Fixed quota-retry filter masking **HttpError** with JSONDecodeError on non-JSON 403s (BigQuery); added a real 403 test.

Projects

NeuraSWE – Autonomous AI Software-Engineering Agent — Code
Python, FastAPI, React, PostgreSQL, Redis, ChromaDB, Docker, Socket.IO, GitHub Actions · 10-tool ReAct agent

- Built a framework-free **ReAct coding agent** (10 tools, no LangChain) that turns GitHub issues and Jira webhook tickets into sandboxed Docker runs: edits code, runs tests, and opens pull requests after re-verifying tests.
- Built **RAG code search**: Python ast chunking (regex fallback elsewhere), nomic-embed-text embeddings, per-repo ChromaDB.
- Built **LLM failover** across Gemini, Groq, OpenRouter, and Ollama (429 cooldown, key rotation, optional hedging), an arq/Redis job queue with Socket.IO streaming, 2 GB / 1 CPU sandbox limits, and ~77 pytest tests in CI.

CodeCollab – Real-Time Collaborative Development Environment — Code | Live
Next.js, Spring Boot, Node.js, Redis, MongoDB, Yjs CRDT, WebRTC, Monaco Editor, Docker

- Architected a multi-file collaborative editor: Monaco and **Yjs CRDT** documents synced over a dedicated Node.js WebSocket service with live cursors and presence, decoupled from the **Spring Boot** REST and STOMP backend.
- Implemented Google/GitHub **OAuth2** with JWT access tokens and httpOnly refresh cookies, service-layer room **RBAC** (owner, editor, viewer), STOMP chat, and **WebRTC** mesh audio, video, and screen sharing with STOMP signaling.
- Built a multi-language code-execution API with **Redis rate limiting** (30 req/min per user) and timeouts, plus a multi-provider AI assistant with fallback; containerized with **Docker Compose** and deployed as a live demo.

FlashFlow – Distributed Flash Sale & Inventory Reservation Engine — Code
Node.js, Express.js, Redis, MongoDB, Docker, Kubernetes, HPA, k6 · 4 microservices

- Built a flash-sale backend as four Node.js **microservices** (API Gateway, Auth, Order, Stock) with gateway-level JWT verification, Redis-backed inventory, and MongoDB persistence for auth and order data.
- Implemented **Redis DECRBY** stock reservation with a negative-result check and INCRBY correction to reject over-subscribed requests; wrote a **k6 stress scenario** (200 VUs, 223K iterations, 2-minute cap).
- Dockerized all services; wrote Kubernetes manifests with NGINX ingress and CPU-based **HPAs** (gateway 1–10, others 1–5).

Technical Skills

Languages: Java, Python, JavaScript, Go, C++, SQL
Backend & Systems: Spring Boot, FastAPI, Node.js, Express.js, Microservices, Distributed Systems, WebSockets, OAuth2
Frontend & Databases: React, Next.js, React Native, Flutter; PostgreSQL, MongoDB, Redis, ChromaDB
Cloud & DevOps: Docker, Kubernetes, AWS, GitHub Actions (CI/CD), Nginx, k6
AI & Fundamentals: LLM Agents (ReAct), RAG, Generative AI; Data Structures and Algorithms, OOP, System Design

Achievements

- LeetCode **Knight** badge with a contest rating of **1939**; solved **600+** problems.
- Merged kubernetes/perf-tests: ClusterLoader2 rolling-update status fix — PR #4182
- Merged kubernetes/website: kubelet reservedSystemCPUs documentation correction — PR #57960
- AWS Academy – Cloud Foundations — Credential
- Oracle Cloud Infrastructure Certified Architect Associate — Certificate
- Oracle OCI Certified AI Foundations Associate; Agentic AI Certified Foundations Associate — AI Certificate | Agentic Certificate